

11/8/26

Applied ML

Machine Learning

- Autonomous driving: object detection
- End to End Learning

Skild.ai → air force robotics

ML Tasks

- ① Classification / Pattern Recognition
- ② Regression / Prediction
- ③ Density Estimation
- ④ Clustering

Quote: P, E.
Performance.

Classifier: - accuracy / Error rate.

- weighted versions of accuracy

Regression: → MSE

→ application dependent.

Density Estimation: log-prob.

* generalisation error.

Experience (E)

Supervised
Unsupervised
Semi-supervised

Supervised + Unsupervised
major focus.

- Transformers are a kind of NN
- Portfolio: supervised learning.
(done by self supervised learning)

feed fwd, perception.

→ Based on Prob stat

- use RL?
- use DT or ensemble?
- use L/USL?

Modelling based techniques

Grading

HW (4 theory + Portfolio) — 20%.

Quiz (Best 3/4) — 20%.

Proj — 15%.

MS — 20%.

SS — 25%.

Q1 — 20 Aug '26
(LA, PnS, Vector
diff)

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Mid - Lec - 2

ML as a fn mapping

$$f(x, w) : \mathbb{R}^n \rightarrow \{ -1, 1 \}$$

Both training & predicting.

Simple Classifier - KNN (Non Parametric)

Regression

Classifier

Problems

hyperparameter

Binary classification (Regression model)

linear classifier : $f(x) = w^T x + b$

learning machine $\hat{y} = f(x; w)$: $w \in \Omega$
choose

Regression

$$R(w) = \iint p(x, y) L(y, f(x, w)) dx dy, w \in \Omega$$

classification :

Gradient descent for Loss minimization :

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Barcode

Calcl

Rank = no. of non-zero rows
Echelon form

ML Tot

Rank

1x, 10x

Numpy

→ np.linspace, arrange.

→ Reshaping & transpose.

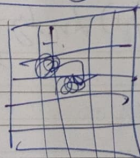
Rank

MML

- orthogonal vectors are $\perp$ → prove it
- Symmetric matrix $\Rightarrow$ eigen values are real
- diff eigen values $\Rightarrow$ eigen vectors $\perp$
- Covariance matrix is the semi-definite & symmetric
- Positive matrix P is idempotent ($P^2 = P$)

Calcl

- 1) Matrix Rank
- 2) 2×2 eigen values & eigenvectors
- 3) Vector Projⁿ on line
- 4) Gram-Schmidt on Vectors
- 5) Bayes rule
- 6) $E(X), Var(X)$



Echelon form

Process: gaussian elim.

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Math for ML

(Tilgater NB will be updated throughout)

→ Binary classification using logistic regression (significance discussed)

Multinomial (Categorical)

Softmax $\rightarrow$ K categories & this is $\rightarrow$ prob over these K categories.

$\pi_k \geq 0, \sum \pi_k = 1$ (closed set of classes)

η = set of parameters

$\eta = (\eta_0, \eta_1)$

Cards (Gaussian/Normal distribution)

$$p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Normalization factor for analog to $\sum()$

clamping $\rightarrow$ shift

clamping $\rightarrow$ flatten

(flatter curve has higher entropy)

Loss fun will be variant of entropy fun only.

→ given the pdf we can find entropy
(measure of randomness)

for fully deterministic system, entropy = 0

$$H(x) = -\sum_k x_k \log x_k$$

So we use it as a loss fun when doing ML models.

for costs distribution $\rightarrow$ cdf

→ sample mean is true mean (estimator)

↑ no of samples the estimate $\rightarrow$ true mean

Conditional Distributions

Bivariate gaussian (jointly gaussian)

$$Y|X = x \sim N(\text{mean}, \text{variance})$$

So we can use d.f. to model. this extends to N dimensional gaussians too.

Probabilistic Model

$$y = f(x_i, w) + \epsilon_i$$

↑ given x_i determine y_i
random error

2. we assume Normal distribution w/ mean of 0.
 $\epsilon_i \sim N(0, \sigma^2)$

→ sell b/c price of house is linear sell

$$y_i = x_i^T w + \epsilon_i$$

$$\downarrow \quad \quad \quad w_1 x_1 + w_2 x_2$$

$$y_i = x_i^T w + \epsilon_i$$

for gaussian σ^2 w/in 25
99% w/in 35

Model will have normally distributed error w/ 0 mean & fixed unknown variance (fixed constant)

x_i : input
 y_i : target

If we write it as conditional distribution,
 $y_i \sim N(\mu_i = x_i^T w, \sigma^2)$

* $H(x)$ to model mean value

$$f(y_i | x_i, w, \sigma^2) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(y_i - x_i^T w)^2}{2\sigma^2}\right)$$

→ For mscn we need to tell target.
What is a notion of w ? So we define
likelihood function.

$$l(w, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - x_i w)^2}{2\sigma^2}\right)$$

& (x_i, y_i) $i=1, \dots, n$ then likelihood is
prod of these densities.

→ this is a joint density for y given x and w .
This quantity shows how likely it is that the observed y values are generated by the model.

$$l(w, \sigma^2 | (x_i, y_i)_{i=1}^n)$$

fixing w & σ^2 will generate error.

It gives a likelihood that error is from this distribution, &

$$y = mx + c \rightarrow c = 0$$

for x , w , c , for every $x_i \rightarrow$ gives y_i

we will have highest density
at the mean.

→ corrections in models for data given it follows that error model.

model's rep of x & y over lines.
this could be representative.

likelihood defines how consistent model is for (x, y)

error → some kind of diff → modeling choice.
whatever probability model we define, for ex, y has max value, model & data are consistent.

We go to maximize → we take log likelihood.

$$l(w, \sigma^2) = \log \left[\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - x_i w)^2}{2\sigma^2}\right) \right]$$

$$= \sum_{i=1}^n \left(-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y_i - x_i w)^2}{2\sigma^2} \right)$$

max of error = max of log likelihood.

{ this is for every parameters for l.s model }

Confusion matrix

2x2

	Actual	
Predicted	TP	FP
	FN	TN

diag

(x, y)
 $P \rightarrow$ Model predicted true
 $N \rightarrow$ " " " false
 $True$ - pred was correct
 $False$ - pred was wrong

$i \in \{1, \dots, N\}$
 TP : predicted + actually
 FP : predicted +, actually -
 TN : predicted -, actually -
 FN : predicted -, actually +

$x_i \rightarrow$ sample
 x_j
 $j \in 1, \dots, d$
 $\vec{x} \in \mathbb{R}^d$
 $\vec{w} \cdot \vec{x} = \sum_{j=0}^d w_j x_j$
 $x_0 = 1$
 $\vec{b} \in \mathbb{R}^d$
 $\vec{b} \cdot \vec{x} \in \mathbb{R}$
 $\vec{b} \cdot \vec{x} \in \mathbb{R}^{d \times d}$
 $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}$

$y \in \{0, 1\}$
 $p \in [0, 1]$
 $\phi = \frac{1}{1 + e^{-\vec{w} \cdot \vec{x}}}$
 invertible mapping from $\vec{w} \cdot \vec{x} \rightarrow \phi$

Linear regression

$$L = \sum_{i=1}^N (y^i - \vec{w}^T \vec{x}^i)^2$$

$$L = \sum_{i=1}^N (y^i - w_0 x_0^i - w_1 x_1^i - \dots - w_d x_d^i)^2$$

$$\frac{\partial L}{\partial w_j} = -2 \sum_{i=1}^N (y^i - \vec{w}^T \vec{x}^i) x_j^i$$

$$\vec{\nabla}_{\vec{w}} L = \begin{bmatrix} \partial L / \partial w_0 \\ \vdots \\ \partial L / \partial w_d \end{bmatrix}$$

$$\vec{w}(t+1) = \vec{w}(t) + \eta \vec{\nabla}_{\vec{w}} L$$

Logistic

$$y \in \{0, 1\}$$

$$\vec{w}^T \vec{x} \in \mathbb{R}$$

$$\downarrow$$

$$f(x, \vec{w})$$

$$\phi = \sigma(\vec{w} \cdot \vec{x})$$

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ML class

linearity $\Rightarrow$ super position + homogeneity

$f(x, w) = w^T x$ is linear in both x and w .

6 weights: $w_0, \dots, w_5$
No bias error.

Small-ish. / ~~test~~ test error.

$$y = ax^2 + bx + c \quad (\text{ideal})$$

but we got 5th degree polynomial in picture
 $\rightarrow$ bumps exactly here.

Simple $\rightarrow$ high bias / underfitting

Complex $\rightarrow$ high variance / overfitting.

Good: just enough.

Big: too much freedom.

Regularization

$$L_2: \text{loss} + \lambda (w_3^2 + w_4^2 + \dots)$$

$\rightarrow$ to penalize large weights for better generalization.

$$\min_w \frac{1}{N} \sum_i (y_i - w^T x_i)^2$$

$$\min_w \left[\frac{1}{N} \sum (y_i - w^T x_i)^2 + \lambda \sum \|w_j\| \right]$$

penalty for
data weights

$$\|w\|_1 = |w_1| + |w_2| + \dots + |w_N|$$

(Lasso Regularization)

$$d_1 / \text{Lasso} = \lambda \sum |w_j|$$

$$d_2 / \text{Ridge} = \lambda \sum w_j^2$$

882.

$$y = w^T x + \text{Gaussian noise. } (\epsilon)$$

"FIND GOOD WEIGHTS"

Linear regression (find the best w)

$$\hat{y} = Xw$$

$$\text{Loss}, d(w) = \sum_{i=1}^N (y^i - w^T x^i)^2$$

$$= \|y - Xw\|_2^2$$

$$= (y - Xw)^T (y - Xw)$$

$$y^T y - y^T X w - w^T X^T y + w^T X^T X w$$

$$y^T X w = w^T X^T y$$

$$d(w) = y^T y - 2w^T X^T y + w^T X^T X w$$

$$\frac{\partial d}{\partial w} = 0 \Rightarrow \hat{w} = (X^T X)^{-1} X^T y$$

(Pseudo inverse / least sq. soln)

* $\neq \lambda w^T w$

$\Rightarrow$ Ridge regression / $\sqrt{2}$ regularization $\Rightarrow$

prob: 5th deep policy $\Rightarrow$ overfitting
 again: don't lose hope etc!

for d error + (w^T penalty)
 (Laplace $\rightarrow$ Laplace) λ_1
 Gaussian $\Rightarrow$ ridge λ_2

$$\nabla d = 2 X^T X w - 2 X^T y + 2 \lambda w$$

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T y$$

likelihood: $P(y | X, w)$

$$P(w) \propto e^{-\lambda w^T w}$$

$$\log P(w) = -\lambda w^T w + C$$

$$w^T w = w_1^2 + w_2^2 + \dots$$

$$= -\lambda \sum w_j^2 + C$$

(A)

n confettors, w weeks

n campsites, w weeks
Every week a list comes out of the Arch

2nd week + 1 part 4/11/21. Score is
cumulative.

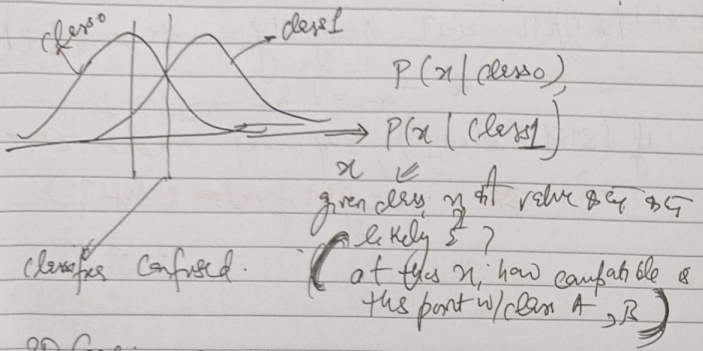
end of week t ,

$$s_{\text{NAK}_i}(t) = 1 + \underbrace{\left[\frac{f}{g_i(t)} - g_i(t) \right]}$$

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ML Class

Probability classifier: class 0 / class 1?



2D Case: 2 features.

$x = (x_1, x_2)$. x_1 ht, x_2 wt.

$$P(y=1 | x_1, x_2)$$

Discriminative / Generative Classifier

~~Discriminative Classifier~~

Logistic Regression is discriminative.

$$P(y=1 | x) = \sigma(w^T x + b)$$

Given x , tell me which class it belongs to.

try to understand y class 0 or 1 data is distributed over x

But $\rightarrow$ Generative classifier learns like $P(x | y)$.

Bayes rule to get $P(y | x)$

So discriminative $\rightarrow$ directly learn $P(y | x)$

Generative $\rightarrow$ learn $P(x | y)$ then Bayes rule to get $P(y | x)$.

$\rightarrow$ Notation: $(x^1, y^1), (x^2, y^2), \dots$

input / features $\rightarrow x$

class $\rightarrow y$

$P(x | y) \Rightarrow$ distribution / prob of x given class y .

If spam answers don't match? $P(x | y=1)$

Non spam $\rightarrow P(x | y=0)$

If spam word $y=1$ free ✓ un ✓ links ✓ $\Rightarrow P(x | y=1)$

Normal $y=0$ week ✓ work ✓ friends $\Rightarrow P(x | y=0)$

Naive email $\rightarrow$ xmas.

Generative Flow $\rightarrow$ $P(x_{new} | y=1)$ is 1 $\rightarrow$ looks spam

$P(x_{new} | y=0)$ is 0 $\rightarrow$ looks Normal.

But we want $P(y | x)$.

given spam likelihood over features.

$$P(y | x) = \frac{P(x | y) \cdot P(y)}{P(x)}$$

(Class conditional distributions)

$\rightarrow$ Prior Probability

$\rightarrow$ Evidence / Marginal Probability

$$P(y|n) = \sigma(wTx + b)$$

88

Neurotic classification (using Bayes classifier)

good product seg. $\max(y|x_1, \dots, x_n)$

88 Rob given this image represents 95 given its pixels.

Pixel 1 = black
2 = white
3 = white
4 = black

Computer has to choose answer $\in \{0, 9\}$
Given these pixels, computer 88 95 & 100 out of 111
And it chooses 9, 10, 0, 9

$P(0|pixels)$, $P(1|pixels)$

Say, 0 $\rightarrow$ 1
1 $\rightarrow$ 2

5 $\rightarrow$ 75%

9 $\rightarrow$ 07.

That is the machine
22 possible answer y of
Rob doesn't give the
input pixels & choose the
maximum

a) how to calculate $P(y|x)$

reverse

$$P(y|x_1, x_2, \dots, x_n) = \frac{P(x_1, \dots, x_n|y) \cdot P(y)}{P(x_1, \dots, x_n)}$$

88 $P(5|x_1, x_2, \dots, x_n)$
Posterior

group digit 5
hardly with image

Digit recognition

$$P(y=5|x) = \frac{P(x|y=5)P(y=5)}{P(x|y=5)P(y=5) + P(x|y=0)P(y=0)}$$

as
if only 2 possibilities

$$P(x) = \text{way 1} + \text{way 2}$$

How many parameters to specify likelihood?
900 pixels together

So there's really pattern A $\rightarrow$ same prob
B $\rightarrow$ "

$\rightarrow$ Likelihood is how do we model $P(x_1, x_2, \dots, x_{900}|y)$

how many parameters are need to specify the likelihood

$$img = 30 \times 30 = 900$$

image 3 pixels, x_1, x_2, x_3

0 $\rightarrow$ white
1 $\rightarrow$ black

Possible images

$$\begin{pmatrix} x_1 & x_2 & x_3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$000 \rightarrow p_7$$

$$111 \rightarrow p_8$$

in 900 pixels, 2^{900} patterns $\rightarrow p_{2^{900}}$
for 4 sig

$$p_1 + p_2 + \dots + p_8 = 1$$

$$p_7 = 1 - p_1 - p_2 - \dots - p_6 \rightarrow 2^n - 1 \text{ ind patterns}$$

FLS. MLE, tons of training data

Naive Bayes

assumption: all features are independent

$$P(x_1, \dots, x_n | y) = \prod_{i=1}^n P(x_i | y)$$

$$P(y | x) \propto P(x | y) \cdot P(y)$$

Training data: frequency of digit.

0	→ 600
1	→ 150
2	→ 50

$$P(y=1) = \frac{150}{1000}$$

$$P(y=2) = \frac{50}{1000}$$

(Prob of each class)

$x_{21} \rightarrow$ row 2 col 1 pixel. $\in \{0, 1\}$

$$P(x_{21}=0 | 1), P(x_{55}=0 | 1)$$

GDA $\Rightarrow$ entire input is for following question:
distribution.

Clustering, means
convex nature
DB scan (density)

Cross Validation $\rightarrow$ Sampling Issue
(Take multiple samples)

Bias & Variance $\rightarrow$ Tradeoff

L_0 norm $\rightarrow$ No. of non zero parameters

$$\vec{w}^{\text{ok}} = \underset{\vec{w}}{\text{argmin}} \sum_{i=1}^N (y^i - \vec{w}^T x^i)^2 + \underbrace{\lambda \|\vec{w}\|_0}_{\substack{\text{Regularization term} \\ \text{Lagrange multiplier} \\ \text{(Complementary slackness)??}}}$$

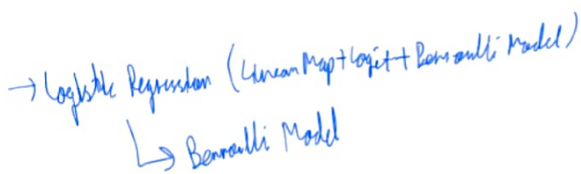
$\lambda \|\vec{w}\|_1$
Sparsity promoting

LASSO

If some w_i 's are 0, $\Rightarrow$ Feature Selection

$$\hat{w} = (X^T X)^{-1} X^T y \Rightarrow \xrightarrow{L_2 \text{ Regularization}} (X^T X + \lambda I)^{-1} X^T y$$

Ridge Regression



Binary cross entropy loss

→ Generative vs Discriminative Probabilistic Classification

$$P(y|x) = \frac{P(x|y) P(y)}{P(x)}$$

↓

Marginal Distribution (Fixed)

$$P(y|x; \theta) = \frac{1}{1 + e^{-\theta^T x}}$$

$(x_1, x_2, \dots, x_{100}) \rightarrow 2(n-1) \text{ parameters} \rightarrow$

↓

$x_1, \dots, x_n$